

Advance Billing

Cloud-Based GST Billing SaaS for Indian Businesses

1. Project Overview

Project Title

Advance Billing – Cloud-Based GST Billing SaaS

Project Type

Software as a Service (SaaS)

Domain

Business Management / Billing & Invoicing

Platform

Web Application

Target Country

India

2. Introduction

Small and medium-sized businesses across India still rely on handwritten invoices, Microsoft Excel, Microsoft Word, or expensive ERP software for their daily billing operations. These approaches are either time-consuming, error-prone, or financially unsuitable for small businesses.

Many existing online billing platforms provide only limited free functionality and require users to purchase subscriptions for basic features. Additionally, several systems are difficult for non-technical business owners to use.

Advance Billing aims to solve this problem by providing a modern, cloud-based, easy-to-use billing solution that automates invoice creation, GST calculations, customer management, and product management while remaining simple enough for any business owner to use.

The primary goal of the project is to reduce manual effort and simplify GST billing without introducing unnecessary ERP complexity.

3. Problem Statement

Small businesses face several challenges while managing invoices:

- Manual invoice creation consumes significant time.
- GST calculations are prone to human error.
- Customer information is scattered across multiple files.
- Product information must be entered repeatedly.
- Previous invoices are difficult to locate.
- Professional invoice templates are often unavailable.
- Existing ERP solutions are expensive and overly complex.
- Many business owners are not comfortable with complicated software.

There is a need for a centralized, cloud-based billing platform that is simple, reliable, and affordable.

4. Proposed Solution

Advance Billing is a cloud-based SaaS platform that enables businesses to:

- Register their organization.
- Verify their email address.
- Manage customers.
- Manage products and services.
- Automatically calculate GST.
- Generate professional invoices.
- Print invoices directly from the browser.
- Generate PDF invoices on demand.
- Download complete business backups.
- Receive periodic backup emails.
- Access business data securely from anywhere.

The application focuses exclusively on billing and intentionally excludes inventory management, accounting, payroll, and other ERP modules.

5. Project Objectives

The primary objectives of Advance Billing are:

- Simplify GST billing.
 - Automate repetitive billing tasks.
 - Eliminate manual calculation errors.
 - Provide a clean and user-friendly interface.
 - Allow businesses to generate professional invoices.
 - Enable easy customer and product management.
 - Ensure data safety through backup mechanisms.
 - Build a scalable SaaS architecture suitable for future expansion.
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6. Target Users

Advance Billing is designed for:

- Retail Stores
- Grocery Shops
- Medical Stores
- Electronics Shops
- Wholesalers
- Manufacturers
- Service Providers
- Freelancers
- Small Businesses
- Medium Businesses

The application is intended for anyone requiring GST invoice generation.

7. Scope of the Project

Included

- User Authentication
- Email Verification
- Organization Management
- Customer Management
- Product Management
- Service Management
- GST Billing
- Invoice Templates

- Browser Printing
- PDF Generation
- Dashboard
- Backup System
- Restore System
- Responsive User Interface

Excluded

- Inventory Management
 - Employee Management
 - Purchase Orders
 - Accounting Ledger
 - Payroll
 - Payment Gateway
 - Barcode Scanning
 - AI Features
 - WhatsApp Integration
 - Mobile Application
 - Multi-country Taxation
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8. System Workflow

Organization visits website.

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Signs Up

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Verifies Email through Brevo

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Logs In

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Completes Organization Setup

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Accesses Dashboard

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Creates Customers

↓

Creates Products / Services

↓

Creates Invoice

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System Calculates GST

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Invoice Preview

↓

Print / Download PDF

↓

Invoice Saved in Database

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Monthly Backup Generated

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Backup Emailed to Organization

9. Authentication Module

Features include:

- Sign Up
- Login
- Email Verification

- Forgot Password
- Reset Password
- Logout

One account represents one organization.

Employee roles are intentionally excluded from Version 1.

10. Organization Module

Each organization maintains:

- Business Name
 - GSTIN
 - PAN
 - Address
 - Phone Number
 - Email Address
 - Logo
 - Letterhead
 - Invoice Prefix
 - Financial Year
 - Currency
 - Default Invoice Template
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11. Customer Module

Customer information includes:

- Customer Name
- GST Number
- Mobile Number
- Email
- Billing Address
- State
- Notes

Duplicate GST numbers should not be allowed.

12. Product & Service Module

Each item contains:

- Product / Service Type
- Name
- HSN/SAC Code
- Unit
- Price
- GST Percentage
- Description

Products and services share the same module.

13. Invoice Module

Each invoice includes:

- Customer
- Invoice Number
- Invoice Date
- Product List
- Quantity
- Unit Price
- Discount
- GST
- Subtotal
- Grand Total
- Notes

The user may edit product prices, GST, quantity, and discounts before finalizing the invoice.

14. GST Calculation

GST calculations are performed exclusively by the backend.

The frontend displays only the calculated values.

Supported GST slabs include all valid Indian GST rates.

The system automatically handles:

- CGST
- SGST
- IGST

according to applicable GST rules.

15. Invoice Templates

The application provides approximately 10–12 professionally designed invoice templates.

Organizations can choose their preferred design.

Only the appearance changes.

Invoice data remains identical.

16. Letterhead Support

Organizations may choose between:

Option A

Standard Invoice Header

or

Option B

Organization Letterhead

If Letterhead Mode is enabled:

- Uploaded letterhead appears at the top.
 - Organization logo is not duplicated.
 - Invoice body uses the selected template.
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17. PDF Generation

Invoices are not permanently stored as PDF files.

The database remains the single source of truth.

Whenever the user selects:

- Print
- Download PDF

the application generates the PDF dynamically.

This reduces storage requirements and simplifies maintenance.

18. Printing Strategy

Invoices are optimized for:

A4 Portrait

Users print directly through the browser.

The browser's built-in "Print → Save as PDF" functionality is supported.

19. Dashboard

Dashboard displays:

- Total Revenue
- Monthly Revenue
- Total Customers
- Total Products
- Total Invoices

Graphs include:

- Monthly Revenue Trend

Recent invoices are also displayed.

20. Backup Strategy

Business data is protected using three layers.

Layer 1

Primary PostgreSQL Database

Layer 2

Manual Backup Download

Users may download:

- Customers
- Products
- Invoices
- Complete Backup

Layer 3

Automatic Monthly Backup

Every month:

- Generate backup
- Compress data
- Email organization
- Delete temporary backup

The application does not permanently store backup archives.

21. Restore Strategy

Organizations can restore previously downloaded backups.

Restore package includes:

- Customers
- Products
- Invoices
- Organization Settings
- Logo
- Letterhead

22. Security

Security measures include:

- Email Verification
- CSRF Protection
- Secure Authentication
- Organization Data Isolation
- Server-side Validation
- Image Validation
- HTTPS Deployment

23. User Interface Principles

The application is designed for users with little technical experience.

Design principles:

- Clean Interface
- Large Buttons
- Minimal Clicks
- Simple Navigation
- Responsive Layout
- Fast Performance

The goal is that even a first-time user can create an invoice without training.

24. Mobile Responsiveness

The application supports:

- Desktop
- Tablet
- Mobile Browser

Invoice creation follows a mobile-friendly step-by-step flow.

25. Technology Stack

Backend

- Django

Frontend

- Django Templates
- HTML
- CSS
- JavaScript

Database

- PostgreSQL

Charts

- Chart.js

Icons

- Lucide Icons

Email

- Brevo

Media Storage

- Cloudinary

Hosting

- Render

Version Control

- GitHub

Static Files

- WhiteNoise
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26. Deployment Strategy

Frontend and backend are deployed together using Django.

Database is hosted on PostgreSQL.

Uploaded media is stored externally.

Email verification and monthly backups are handled through Brevo.

27. Future Enhancements

Future versions may include:

- Multi-user organizations
 - Subscription plans
 - Inventory Management
 - Accounting
 - Payment Gateway
 - API Access
 - Mobile Application
 - AI Features
 - Analytics Dashboard
 - GST Return Assistance
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28. Conclusion

Advance Billing is designed as a modern cloud-based GST billing platform focused on simplicity, automation, and reliability. By combining an intuitive interface, professional invoice generation, automated GST calculations, and dependable backup mechanisms, the system enables businesses to manage their billing process efficiently without the complexity of traditional ERP software.

The architecture has been intentionally designed to support future expansion while keeping Version 1 focused on solving the core billing problem for Indian businesses.